

سکے الیہ

## Spectrum

The ordered distribution of EM waves according to ① frequency and ② wavelength.

Physical properties are determined by their wavelength.

الطيف الكهرومغناطيسي

# Spectrum

harmfull

## ① Radio waves

Radio waves are generated by LC oscillators

they are used in

- ① Radio, ② TV ③ Communication

## ② Micro waves.

Microwaves are short waves

they are used in ① Radar ② Oven ③ Communication

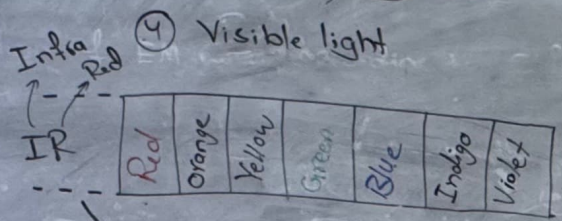
## ③ Infra Red:

IR is produced from hot bodies

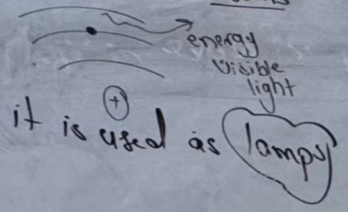
they are used in ① Mobile phone ② Remote control.

it appears as heat

- ③ Communication

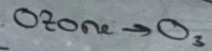


The very small part of spectrum that can be seen by human eyes + is produced by rearrangement of electrons in atoms



## ⑤ Ultra Violet → UV

the mainly source is Sun



This layer lies in upper atmosphere layer is called stratosphere



UV

Ozone converts UV into IR and warms the air.

converter

IR

## ⑥ X-ray

X-ray is produced from deceleration of high speed electrons.

they are used in ① medical field ② Airport

Gamma ray is produced from a nucleus

it causes a damage for living tissues (high penetrating)

- ① damage
- ② medical field

EM wave